Ayush Mohanty — Curriculum Vitae

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EDUCATION

- **Ph.D. Candidate in Machine Learning (School of Industrial & Systems Engineering)**
  Georgia Institute of Technology, Atlanta, GA — *2020-2026*
  - **Advisor:** Nagi Gebraeel
  - **Doctoral Minor:** System Informatics and Control
- **M.S. in Computer Science (Specialization: Machine Learning)**
  Georgia Institute of Technology, Atlanta, GA — *2024-2025*
- **B.S. in Manufacturing Sci. & Engg. with M.S. in Industrial Engg. & Mgmt. (Dual Degree)**
  Indian Institute of Technology (IIT) Kharagpur, India — *2015-2020*

RESEARCH INTERESTS

Methodology: Data Science, Federated Learning, Causal Reasoning, Fault Diagnosis & Prognostics, State Estimation & Control
Applications: Cyber-Physical Systems (Manufacturing, Aerospace, and Power Systems), Internet-of-Things, Robotic Manipulator

PEER-REVIEWED PUBLICATIONS

[1] *A. Mohanty*[#], *J. Dekarske, S. Joshi, S. Robinson, N. Gebraeel*, **"Prognostic Framework for Robotic Manipulators Operating Under Dynamic Task Severities"**, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 56, no. 1, pp. 443-457, Jan. 2026 Paper

- **Top publication venue for Automation & Control Theory** – Google Scholar Metrics.

[2] *A. Mohanty*[+#], *N. Mohamed*[+], *P. Ramanan, N. Gebraeel*, **"Federated Granger Causality Learning For Interdependent Clients With State Space Representation"**, *International Conference on Learning Representations (ICLR) 2025* Paper

- **CORE A* – flagship venue, top conference in ML and AI (20-25% acceptance rate)** – Google Scholar Metrics.

[3] *S. Sherman, P. Pischulti, A. Mohanty, M. Hwang, D. Ivey, S. Robinson, M. Berges, N. Gebraeel, D. Klaus, A. Anderson*, **"A Development Framework for a Comprehensive Capstone which Demonstrates Human Interaction with Autonomous Habitat Technology"**, *International Conference on Environmental Systems, 2024* Paper

[4] *D. Ivey, T. Barkouki, M. Torralba, U. Ulusoy, S. Eshima, A. Mohanty, C. Lindbeck, S. Balakirsky, S. Robinson*, **"Design, Build, Test of a CO2 Removal Testbed and Twin Robotically Manipulable Testbed: Sensing Degradation and Performing Maintenance with Robot/Human Teaming"**, *International Conference on Environmental Systems, 2023* Paper

[5] *S. Eshima, J. Nabity, A. Mohanty, H. Rozas, N. Gebraeel*, **"A Diagnostics Model for Detecting Leak Severity in a Regenerable CO2 Removal System"**, *International Conference on Environmental Systems, 2022* Paper

[6] *S. Singh, A. Mohanty, R. Rai, B. Mahanty, M.K. Tiwari*, **"An Optimization Framework for Operational-Level Resource Composition in an Inclusive Manufacturing System"**, *ASME Journal of Computing and Information Science in Engineering (JCISE), October 2022; 22(5): 051003* Paper

[7] *A. Rudrapal, A. Mohanty, D.K. Pratihar*, **"Improvement of Crow Search and Bat Algorithms Using Two Novel Mutation Operators"**, *International Conference on Computers & Industrial Engineering (CIE), 2018* Paper

[8] *A. Mohanty*[#], *J. Sindha, A. Panda, D. Chakravarty*, **"A Novel Approach for Optimal Speed Control of the Vehicle Using Drive Envelope Based Analysis"**, *IEEE Transportation Electrification Conference (ITEC-India), pp. 1-5. IEEE, 2017.* Paper

PREPRINTS AND UNDER REVIEW

[9] *A. Mohanty*[#], *N. Mohamed, N. Gebraeel*, **"Uncertainty in Federated Granger Causality: From Origins to Systemic Consequences"**, Paper

[10] *A. Mohanty*[#], *P. Ramanan, N. Gebraeel*, **"Learning Unknown Interdependencies for Decentralized Root Cause Analysis in Nonlinear Dynamical Systems"**, Paper

[11] *N. Mohamed*[+], *A. Mohanty*[+,#], *N. Gebraeel*, **"Federated Causal Representation Learning in State-Space Systems for Decentralized Counterfactual Reasoning"**, Paper

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[+]Equal contribution [#]Corresponding author

[12] *A. Tursucular, A. Mohanty#, N. Mohamed, N. Gebraeel*, **"Federated Learning of Nonlinear Temporal Dynamics with Graph Attention-based Cross-Client Interpretability"**, Paper

[13] *B. Peters, A. Mohanty#, X. Fang, S. Robinson, N. Gebraeel*, **"Prognostics for Autonomous Deep-Space Habitat Health Management under Multiple Unknown Failure Modes"**, Paper

- **Previous version of the preprint cited by a publication in** *Reliability Engineering & System Safety*

MANUSCRIPTS IN PREPARATION

[14] *A. Mohanty, N. Gebraeel*, **"Federated Learning of Rate Interactions in Coupled Stochastic Differential Equations"**,

[15] *A. Mohanty, N. Mohamed, N. Gebraeel*, **"Cyberattack Detection and Localization in Interconnected Power Systems via Federated Granger-Causal Analysis"**,

[16] *N. Mohamed, A. Mohanty, N. Gebraeel*, **"Federated Kalman Filtering for Networked Systems with Unknown Dynamics"**,

LEADERSHIP AND RESEARCH PROTOTYPING

1. **Capstone: Decentralized Root Cause Analysis & Repair in a Deep Space Habitat** (NASA HOME STRI, 2024)
   - **Role:** *Capstone Captain*; presenter of the Interdependency Learning Framework.
   - **Stakeholder Engagement:** NASA Ames Research Center, NASA Johnson Space Center, Collins Aerospace.
   (a) Live-demonstrated a decentralized machine learning framework for fault detection and root cause analysis across interdependent subsystems, using HOME mockup, and STEVE testbed (both present at Aerospace department, CU Boulder).
   (b) Highlighted cascading fault scenarios and the framework's ability to autonomously trace the root subsystem in real time.
   (c) Integrated root cause analysis results to inform the crew for repair. The crew's cognitive state data was used to dynamically model repair decision-making during the live session.
2. **Scenario Demonstration: Decentralized Orchestration of Edge Analytics in Smart Habitats** (NASA HOME STRI, 2023)
   - **Role:** *Scenario Scientist*; host of the live demonstration and lead for the Robotic Edge Computer.
   - **Stakeholder Engagement:** Blue Origin, NASA Ames Research Center, NASA Johnson Space Center, KBR Inc.
   (a) Demonstrated a real-time orchestration of predictive analytics across four edge computers (representing habitat subsystems) and a central Vehicle System Manager (VSM) in front of NASA and industry stakeholders.
   (b) Deployed Docker-based algorithms autonomously via gustavo, with each edge computer processing its own local data to generate subsystem-specific insights such as anomaly detection, remaining useful life, fault diagnosis, and overheating.
   (c) Showed how insights flowed back to the VSM to drive real-time optimization for repair scheduling using gustavo comms

TEACHING EXPERIENCE

- **INSTRUCTOR**

1. **Fall 2025:** ISyE 3770, Statistics & Applications, Georgia Tech
   Class Size: 75 students
   Duties: Was responsible for the entire course, including preparing syllabus, designing lectures, creating new homework problems, writing quiz and exam questions, and holding instructor office hours.
2. **Spring 2026:** ISyE 3030, Basic Statistics, Georgia Tech (Guest instructor)
   Class Size: 14 students
   Duties: Taught multiple lectures.

- **TEACHING ASSISTANT**

3. **Spring 2026:** ISyE 6381, Manufacturing Reliability, Georgia Tech
   Duties: Graded assignments and exams, and held a weekly lecture to guide students through homework exercises.
   - Taught "*Reliability & survival analysis*" module of JMP.
4. **Fall 2023:** ISyE 6381, Manufacturing Reliability, Georgia Tech
   Duties: Graded assignments and exams, and held a weekly lecture to guide students through homework exercises.
5. **Fall 2021:** ISyE 4803/6805, Reliability Engineering, Georgia Tech
   Duties: Conducted a lecture to discuss midterm solutions, graded homework and exams, and held office hours.
6. **Fall 2021:** ISyE 6381, Manufacturing Reliability, Georgia Tech
   Duties: Graded assignments and exams, and held a weekly lecture to guide students through homework exercises.

SERVICE AND OUTREACH EXPERIENCE

PEER-REVIEWING

- **2026-Present:**

1. Transactions on Machine Learning Research (TMLR) – *Reviewer*
2. International Conference on Machine Learning (ICML) – *Program Committee Member*
3. IEEE Transactions on Automation Science and Engineering – *Reviewer*
4. Communications in Statistics: Simulation and Computation – *Invited by the Editor*

- **2025-2026:**

5. Annual Conference on Artificial Intelligence and Statistics (AISTATS) – *Program Committee Member*
6. International Conference on Learning Representations (ICLR) – *Program Committee Member*
7. AAAI Conference on Artificial Intelligence (AAAI) – *Program Committee Member*
8. Communications in Statistics: Simulation and Computation – *Invited by the Editor*
9. Conference on Neural Information Processing Systems (NeurIPS) – *Subreviewer (invited by advisor)*
10. ACM Practice and Experience in Advanced Research Computing (PEARC) Conference – *Program Committee Member*

- **2024-2025:**

11. International Conference on Learning Representations (ICLR) – *Program Committee Member*
12. Annual Conference of the Prognostics and Health Management (PHM) Society – *Guest Reviewer*
13. IEEE Transactions on Industrial Electronics – *Subreviewer (invited by advisor)*

- **2023-2024:**

14. INFORMS Manufacturing & Service Operations Management Journal – *Subreviewer (invited by advisor)*
15. Technometrics – *Subreviewer (invited by advisor)*

MENTORING

1. **2024-Present:** *Nazal Mohamed* – 3rd Year Ph.D. Student
   - Mentored on the art of defining research problems and thinking clearly through research
   - Provided extensive guidance on academic writing and research paper preparation
   - Collaborated on 3 manuscripts submitted to machine learning conferences. Currently collaborating on 2 journal manuscripts.
2. **2025-Present:** *Ayse Tursucular* – 2nd Year Ph.D. Student
   - Guided her in defining a research problem and deriving mathematical proofs
   - Mentored on research paper writing and effective presentation of results
   - Collaborated on 1 research paper, currently doing joint work on the second manuscript.

COMMUNITY INVOLVEMENT

- **2025:** Judge for K-12 InVenture Prize State Finals (Georgia)

STUDENT ACTIVITIES

- **2018-2019:** *Mechanical Team Lead* of the runner's up team (twice) in Intelligent Ground Vehicle Competition (IGVC)
- **2016-2018:** *Public Relations Head* of Asia's largest AstroTech fest – National Students' Space Challenge (NSSC)

HONORS AND AWARDS

1. **Travel Award Grant, ASA QPRC Conference** (2025)
   Awarded a full travel grant by the American Statistical Association (ASA) & the National Science Foundation (NSF) to present research at the Quality and Productivity Research Conference.
2. **AI Safety Fellowship** (Spring 2025)
   Selected as a fellow in the Georgia Tech AI Safety Initiative. Participated in a 6-week seminar with reading group discussions on *Reward Misspecification and Instrumental Convergence*; *Goal Misgeneralization*; *Task Decomposition for Scalable Oversight*; *Adversarial Techniques for Scalable Oversight*; *Interpretability*; *Criticisms and Whitepaper* in Large AI Models.
3. **Novelis Graduate Scholarship** (2024–2025)
   Recipient of an $8,000 fellowship awarded by the Novelis Innovation Hub at Georgia Tech. Culminated with a thesis presentation at Novelis' Global Research and Technology Center (NGRTC) in Kennesaw, Georgia.

4. **Silver Medal, Department Rank 1, IIT Kharagpur** (2020)
   Awarded the silver medal for securing the top rank in the Manufacturing Science & Engineering dual degree (B.S. + M.S.) program at IIT Kharagpur.
5. **Institute Proficiency Award, IIT Kharagpur** (2019)
   Recognized with the institute-wide award for the best Bachelor's project in Manufacturing Science & Engineering.
6. **Winners, Smart India Hackathon** (2019)
   National-level recognition as winners of the 3rd edition of the Ministry of Human Resource Development's Smart India Hackathon, competing among teams from across the country.
7. **Change of Major Opportunity, IIT Kharagpur** (2016)
   Earned an opportunity for changing major into Manufacturing Science & Engineering by placing within the top 7% of a cohort of over 1300 students after the first year of study.

INTERNSHIPS AND RESEARCH PROJECTS

1. **Prognostics in Hydraulic Turbine Components [Certificate]** **Argonne National Lab, USA**
   Research Intern – **Advisors/Collaborators:** *Dr. Feng Qiu*, *Dr. Murat Yildrim* *Summers 2023*
   • Developed LSTM-based degradation detection and Bayesian exponential degradation models for Remaining Useful Life (RUL) prediction in hydraulic turbines using real-time data streaming *Hydropower Research Institute* (HRI)
2. **Industrial Optimization for Inclusive Manufacturing Framework** **IIT Kharagpur, India**
   Research Assistant — **Advisor:** *Prof. Manoj Kumar Tiwari* *Sep 2019 – Sep 2020*
   • Formulated multi-objective constrained optimization and solved with chaotic MOPSO for cost- and emission-aware manufacturing planning. The work is published in the ASME *Journal of Computing and Information Science in Engineering*.
3. **Metrological Investigation of Industrial X-ray CT Data [Certificate]** **Warwick Manufacturing Group, England**
   Visiting Researcher — **Advisor:** *Prof. Mark A. Williams* *Summers 2019*
   • Designed histogram-based CT data characterization and implemented image processing algorithms, achieving noise reduction in X-ray CT images of a metallic artifact.
4. **Intelligent Racking System Design for Aquaponics Industry [Certificate]** **University of Alberta, Canada**
   Visiting Research Assistant — **Advisor:** *Prof. Rafiq Ahmad* *Summers 2018*
   • Designed energy-efficient, Industry 4.0-aligned intelligent racking system prototypes for aquaponics production.
5. **Parameters Estimation of Quadcopters & Kinodynamic Path Planning [Certificate]** **MathWorks, India**
   Project Intern — **Supervisor:** *Dr. Venkatesh B. Subburaman* *Summers 2017*
   • Wrote a white paper to estimate black-box quadcopter parameters for a MATLAB toolbox
   • Implemented kinodynamic RRT* algorithm for 3D UAV motion planning.

KEY PRESENTATIONS

1. **Interpretable Causal Models for Cascading Failures in Distributed Manufacturing Networks**
   Novelis Global Research & Technology Center, Kennesaw 2025
2. **Federated Causal Reasoning for Dynamic Systems with Interdependent Clients**
   ASA Quality and Productivity Research Conference, Seattle 2025
3. **Learning Unknown Interdependencies in Distributed Manufacturing**
   Thesis Pitch Competition, IISE Doctoral Colloquium, Atlanta 2025
4. **Federated Granger Causality Learning For Interdependent Clients With State Space Representation**
   International Conference on Learning Representations (ICLR), Singapore 2025
5. **Federated Anomaly Detection in Vertically Partitioned Datasets**
   INFORMS Annual Meeting, Seattle, 2024
6. **A Two-Stage Prognostics Framework for Guide Bearing Failures in Hydropower Operations**
   CEESA Seminar, Argonne National Laboratory, Chicago, 2023
7. **Condition-based Task Scheduling for Self-Aware Robotic Manipulators**
   IISE Annual Conference, New Orleans, 2023
8. **Operational Self-Awareness of Robotic Manipulators Performing Tasks with Time-Varying Severity Levels**
   INFORMS Annual Meeting, Indianapolis, 2022
9. **Operational Self-Awareness of Robotic Manipulators Performing Tasks with Dynamic Severity Levels**
   International Workshop on Interdisciplinary Paradigms for Semi-Autonomous Deep-Space Habitation, San Antonio, 2022

10. **A Diagnostics Model for Detecting Leak Severity in a Regenerable CO2 Removal System**
    International Conference on Environmental Systems (ICES), St. Paul, 2022
11. **Improvement of Crow Search and Bat Algorithms Using Two Novel Mutation Operators**
    International Conference on Computers & Industrial Engineering, Auckland, 2018
12. **A Novel Approach for Optimal Speed Control of the Vehicle Using Drive Envelope Based Analysis**
    IEEE International Transportation Electrification Conference, Pune, 2017

REFERENCES

1. **Nagi Gebraeel** (Thesis Advisor),
   Georgia Power Early Career Professor and Professor,
   School of Industrial & Systems Engineering,
   Georgia Institute of Technology
   Contact: nagi.gebraeel@isye.gatech.edu

2. **Stephen K. Robinson** (Director of NASA HOME STRI, PhD Thesis Committee Member, and Collaborator)
   NASA Astronaut (retired) & Professor,
   Department of Mechanical and Aerospace Engineering,
   University of California Davis
   Contact: stephen.k.robinson@ucdavis.edu

3. **Sanjay Joshi** (PI in NASA HOME STRI and Collaborator)
   Professor,
   Department of Mechanical and Aerospace Engineering,
   University of California Davis
   Contact: maejoshi@ucdavis.edu

4. **Pascal Van Hentenryck** (PhD Thesis Committee Member),
   A. Russell Chandler III Chair and Professor,
   School of Industrial & Systems Engineering,
   Georgia Institute of Technology
   Contact: pvh@gatech.edu

5. **Paritosh Ramanan** (Collaborator and Mentor),
   Assistant Professor,
   Industrial Engineering and Management,
   Oklahoma State University
   Contact: paritosh.ramanan@okstate.edu